

Jiaqi Ge

Email: ge24876@gtit.edu.cn

EDUCATION

Guangdong Technion – Israel Institute of Technology (English-taught Program)

B.Sc. in Mathematics with Computer Science (GPA: 89.7/100)

Sep 2023 – Present

PUBLICATIONS

Know Before Fix: Question-Driven Repository Knowledge Acquisition for Software Issue Resolution

Co-author, under submission to ASE

WORK EXPERIENCE

Undergraduate Teaching Assistant – Linear Algebra & Calculus

Shantou, China

Guangdong Technion – Israel Institute of Technology

Mar 2026 – Present

- Facilitate student learning in weekly workshops by providing guidance on complex mathematical problems and theorems.
- Assist students in mastering foundational concepts of Linear Algebra and Calculus through interactive problem-solving sessions.

Undergraduate Teaching Assistant – Python

Shantou, China

Guangdong Technion – Israel Institute of Technology

Oct 2025 – Jan 2026

- Assisted in teaching the introductory Python course and supervised lab sessions.
- Supported students with debugging code and understanding core programming concepts.

RESEARCH EXPERIENCE

Research Assistant – Know Before Fix: Issue-Driven Repository QA for Software Issue Resolution

Shanghai Jiao Tong University, advised by Xiaodong Gu

Feb 2026 – Present

- Contributed to the development and evaluation of **ACQUIRE**, an issue-driven repository QA module that enables software agents to proactively acquire repository-specific knowledge before patch generation.
- Designed and executed comparative experiments on SWE-bench-style tasks, benchmarking **ACQUIRE**-augmented agents against baselines to evaluate question quality and trajectory stages.
- Built comprehensive analysis pipelines to evaluate **ACQUIRE** at question-level, set-level, and pairwise granularities; measured its impact on repair success, patch alignment, and stage-wise behavior.
- Orchestrated result interpretation and manuscript preparation, including the design of ablation studies and the creation of technical figures to demonstrate **ACQUIRE**'s effectiveness.

Research Assistant – Explaining AI Models through Metamorphic Property Generation

Advisor: Prof. Nazareno Aguirre nazareno.aguirre@gtit.edu.cn

Oct 2025 – Feb 2026

- Built a grammar-based fuzzing pipeline to generate multi-step text perturbation chains (typos/punctuation/entity swaps/negation.....) using Checklist .
- Implemented candidate metamorphic-property generation and checking (INV invariance / DIR directional) to evaluate model behavior under k-step transformations.

- Integrated a HuggingFace sentiment-analysis model with Checklist to run automated INV/DIR test suites at scale; summarized experimental findings for research documentation.

Research Assistant – Numerical Methods for Boundary Value Problems((Project Score: 95/100)

Advisor: Prof. Antti Rasila antti.rasila@gtit.edu.cn

Aug 2025 – Nov 2025

- Conducted research on numerical solutions for boundary value problems (BVPs) in MATLAB.
- Reviewed classical theory including existence/uniqueness, Green’s functions, and Sturm–Liouville problems.
- Compared shooting, finite-difference, and spectral methods for linear and nonlinear BVPs.
- Used MATLAB Chebfun toolbox to solve nonlinear systems, singular Sturm–Liouville problems, and chaotic dynamical systems.
- Authored a 20-page report titled “Solving Boundary Value Problems Using Chebfun”.

Research Assistant – Integration of Multi-Age Clocks for Alzheimer’s Disease Diagnosis

Advisor: Dr. Shibiao Wan swan@unmc.edu, University of Nebraska Medical Center (UNMC) May 2024 – Jun 2025

- Integrated brain-aging and epigenetic aging models to support Alzheimer’s disease (AD) diagnosis, and analyzed relationships among different age clocks for age-related disease prediction.
- Contributed to the implementation and code review of a multi-modal age-clock integration framework combining DNA methylation and neuroimaging data.
- Conducted literature reviews on biological aging biomarkers and age estimation models.

SKILLS

Programming: Python, C, C++, Java, MATLAB, Assembly

Languages: Chinese (Native), English (Fluent)

AWARDS

Scholarship – 28,500 USD (2023)

Awarded by Guangdong Technion – Israel Institute of Technology for performance in the Chinese National College Entrance Examination (Gaokao).